Waipapa Taumata Rau
University of Auckland

Part IV Project #123
Department of Electrical, Computer, and Software Engineering

AI-Based Virtual Parliament

Cooperative debate and hidden adversarial strategies

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We built a virtual New Zealand Parliament with cooperative and adversarial AI MPs. Each follows a party persona, and selected agents get private disruption tactics. The latest study tests whether Hansard grounding improves debate quality.

1 One platform, three interfaces

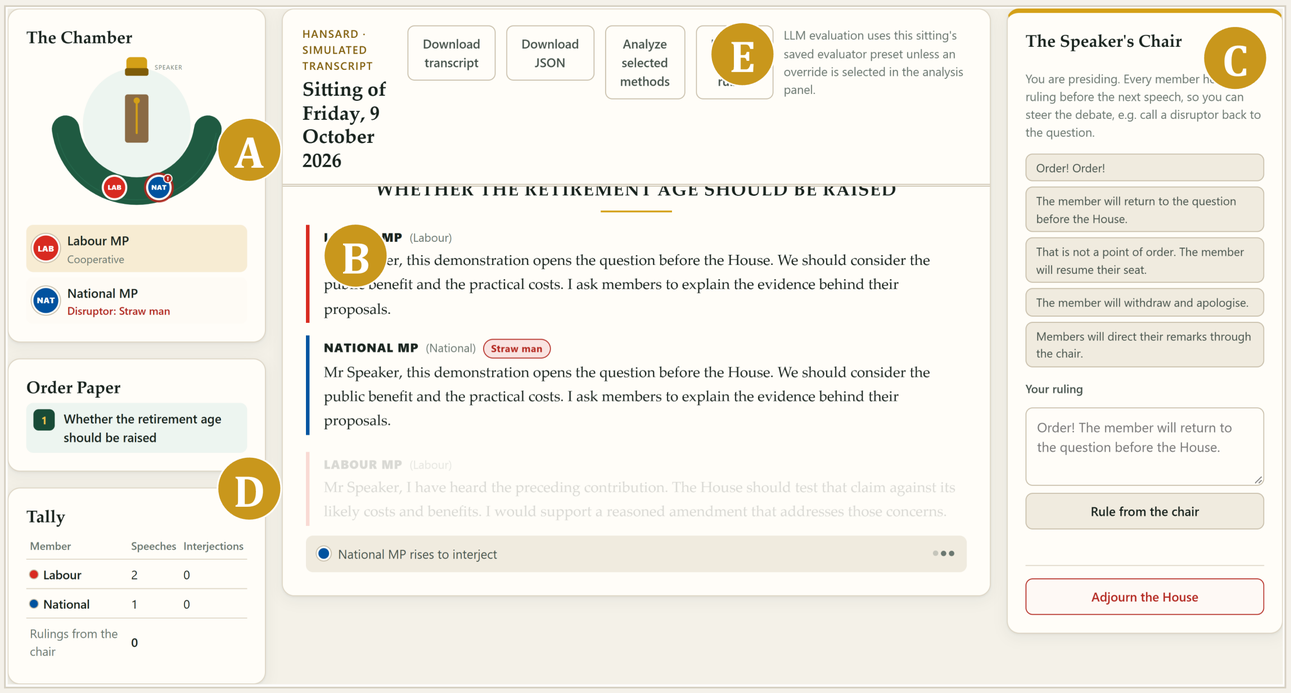


Fig. 1 Deterministic demo: National has a straw-man assignment. Browser, guided CLI and commands share one backend.

A The Chamber
One MP represents each chosen party.
Speakers light up; adversarial roles appear in red.

B Live Hansard
Speeches and interjections stream into Hansard.
Saved sittings can be replayed after a restart.

C The Speaker's Chair
Chair rulings enter the shared public record and reach every MP before the next speech.

D Order Paper and Tally
The agenda advances through topics in order.
Live totals track speeches, interjections and rulings.

E Exports and evaluation
Download public JSON or text, then run local sentiment/stance analysis or separate rubric scores.

2 Cooperative and adversarial agents

Real Hansard
500 reviewed excerpts, 100 per party, from real NZ speeches. The study uses 0, 10 or 100 per party, selected in fixed corpus order [1].

→

Private setup
Each MP gets its own persona, grounding and any hidden tactic. Editable prompts and settings freeze at run start. Other MPs cannot inspect this setup.

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MP agents
Cloud or local models represent selected parties. Cooperative MPs argue policy. **Adversarial MPs** privately receive a tactic to attempt disruption.

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Debate engine
One scheduler manages turns, topics, interjections and chair rulings. Public claim references and each MP's position history guide replies and concessions.

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Saved evidence
Save public speeches and chair events with stable IDs. Replay/export transcripts and separate reports while keeping private prompts and credentials out.

Topic derailment
Steer discussion away from the agenda without acknowledging it.

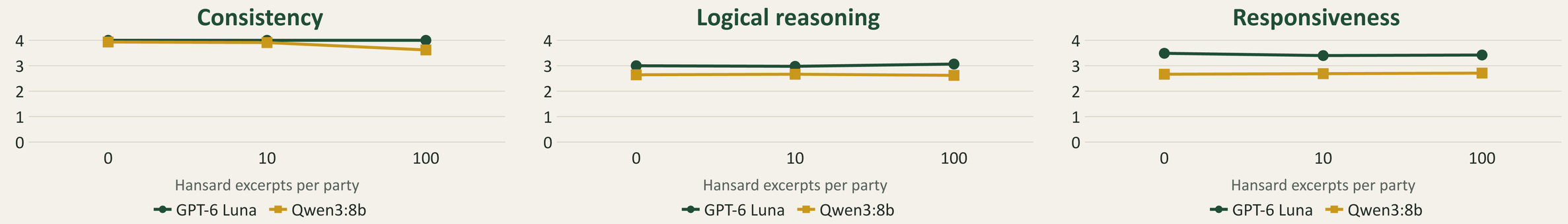
Straw man
Misrepresent an opponent's claim, then attack the distorted version.

Procedural manipulation
Exploit rules or points of order to obstruct substantive debate.

Only assigned MPs see their tactic. Other agents receive their own setup and public evidence, with no notice of hidden assignments.
Select debate and judge models independently. Five rubric metrics, VADER/Cardiff sentiment and DeBERTa stance stay separate.

3 More grounding gave no consistent gain

54 baseline debates and 1,350 speeches [2]. GPT-6 Luna and Qwen3:8b each ran three topics × three grounding levels (0/10/100) × three repetitions, with five parties and five rounds. Topics: retirement age, nuclear power and a four-day workweek. **All agents were cooperative**, with interruptions disabled.



Qwen consistency fell from 3.93 to 3.62 (-0.31) between 0 and 100 excerpts. Other changes were small or mixed. Relevance endpoints were unchanged (Luna 4.00, Qwen 3.98), and disruptive rhetoric stayed near zero (0.00-0.04). Higher rhetoric means more disruption.

Means pool 9 debates per model/level and 45 party scores per metric. One common GPT-6 Luna judge rates public evidence. Scores within a debate are dependent, and automated ratings lack human validation. These baseline runs do not establish adversarial robustness.

4 Outcomes and next steps

Platform: Private personas, hidden tactics and frozen settings support reproducible comparisons. Extra Hansard showed no consistent score gain in the baseline study.

Next: Test each hidden tactic against cooperative baselines. Expand topics and repetitions, and use human reviewers to assess disruption, reasoning and concessions.

Albert Sun: frontend, live chamber, initial CLI and Hansard grounding.
Kieran Joe: architecture, agent engine and supervisor coordination.

References

[1] Rauh & Schwalbach (2020). ParlSpeech V2.
[2] Project 123 debate evaluation study, October 2026.

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Political science collaborators, Georgetown University.
Members are AI archetypes, not real MPs.

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